XTSF: Synthetic Forecasting Explanations with XPC

Executive Summary

Thesis project

Results available through 7 August 2026

Decision snapshot

Interventional Shapley converges toward the known additive structural contribution. Conditional Shapley also converges toward its own exact estimand, but retains a large structural mismatch because dependent features redistribute credit. Coalitional grouping gives the best current accuracy/cost compromise; the simplified game is fastest but changes the estimand materially.

1 Convergence

Increasing coalition or masking budgets reduces Monte Carlo error for both methods. At 32 sampled coalitions, interventional MSE is 0.103 against both its exact estimand and structural truth. Conditional MSE is 0.172 against its exact estimand but 7.31 against structural truth. With 32 masking samples, the corresponding values are 0.088 for interventional, and 0.333 versus 7.53 for conditional.

The persistent conditional structural error is therefore not primarily a Monte Carlo failure. It is a difference between the conditional attribution estimand and the chosen signed structural decomposition.

2 Aggregation experiment

Method	Mode	Structural MSE	Estimand MSE	ms/point	Rows
Interventional	coalitional	0.171	0.171	4.14	1024
Interventional	default	0.512	0.512	12.40	3072
Interventional	simplified	1.594	1.594	0.18	64
Conditional	coalitional	10.72	0.573	22.31	1024
Conditional	default	12.48	1.362	48.15	3072
Conditional	simplified	11.57	1.267	0.79	64

Coalitional grouping improves accuracy over post-hoc default aggregation while using one third as many model rows. The two-player simplified mode is orders of magnitude faster but has noticeably larger estimand error and a different interaction allocation.

3 Supported conclusions

- Monte Carlo convergence must be reported separately from estimand validity.
- Exact conditional explanations are not automatically structural explanations under feature dependence.
- Grouping before explanation is preferable to post-hoc aggregation in the current synthetic benchmark.
- Simplified games are useful speed controls, not drop-in equivalents.

4 Limitations and next evidence

The evidence is synthetic and model-specific. Fixed output filenames and a cached panel require disciplined regeneration. The next study should vary dependence strength, background size, model class, and grouping structure while retaining exact-estimand and structural-error diagnostics.